

Assignment # 1

26-0048

Fiza

Part A- foundational practice problems

Q1:

Algorithm

start

Input N

Set $i = N$

while $i \geq 1$

Print i

$i = i - 1$

stop

pseudocode

Begin

Input N

for $i = N$ Down to 1

print i

End for

End

Q2:

Algorithm

Start

Input year

If $(\text{year} \% 4 == 0 \text{ AND } \text{year} \% 100 \neq 0) \text{ OR } (\text{year} \% 400 == 0)$

Print Leap year

Else

Print Not Leap year

Stop.

Pseudocode

Begin

Input year

If $(\text{year MOD } 4 = 0 \text{ AND } \text{year MOD } 100 \neq 0) \text{ OR } (\text{year MOD } 400 = 0)$

Print "Leap year"

Else

Print "Not Leap year"

End if

End

Q3:

Algorithm

Start

Input Num

Set $\text{count} = 0$, $\text{temp} = \text{Num}$

while $\text{temp} \neq 0$

$\text{temp} = \text{temp} / 10$

$\text{count} = \text{count} + 1$

print count

stop

Pseudo code

Begin

Input Num

$\text{Count} = 0$

$\text{temp} = \text{Num}$

While $\text{temp} \neq 0$

$\text{temp} = \text{temp} / 10$

$\text{Count} = \text{Count} + 1$

End while

Print "Digits": , Count

End

Q4:

Algorithm

Start

for $i = 1$ to 10

If $i \% 3 == 0 \rightarrow$ skip to next

print Table of i

For $j = 1$ to 10

print $i \times j = i * j$

Stop

Pseudocode

Begin

Set $i = 1$ to 10

if $i \text{ MOD } 3 = 0$ then

Continue

End if

Print "table of i ", i

For $j = 1$ to 10

Print $i \times j = i * j$

End for

End

Qs:

Algorithm

Start

Input num

while num > 9

Set sum = 0

while num != 0

sum = sum + num % 10

num = num / 10

print "Intermediate sum:", sum

num = sum

print "final digit:", num

stop.

Pseudocode

Begin

Input num

while Num > 9

sum = 0

while num ≠ 0

sum = sum + (num Mod 10)

num = num / 10

End while

Print "Intermediate sum:", sum

num = sum

End while

Print "final single digit:", num

End

Part B System design scenarios

Q1- Hostel booking system

Algorithm

Start

Input season "peak" or "off peak"

Input Room type "standard", "deluxe", "suite"

Input Nights $\leftarrow$ number of nights stayed

If season = peak then

 if room type standard then
 rate = 5000

Else if room type = deluxe then
 rate = 8000

Else

 rate = 12000

end if

Else

 If room type = standard Then
 rate = 3000

Else if room type = Deluxe then
 rate = 5000

Else

 rate = 800

end if

end if

total = rate * nights

if nights > 7 Then
discount = total * 0.15
total = total - discount
else

discount = 0
print total, discount
Hostel total revenue = hotel total revenue +
total
stop .

2. IPO chart:

Input
season

Process

Output

Solution alternative

check season
and room-type
with nested
if-else to
get rate

Guest final
price

Instead of
nested if-else,
Use a 2D rate
table/array.

room-type

nights

Calculate total =
rate * nights
check if nights
> 7 then
total = total -
15% of total

Discount
amount

Hotel total
Revenue

3. PAC Chart:

Problem

Input

Process

Output

Calculate
hotel bill
for 1 guest
with season
and room-type
Apply 15%
discount if
stay > 7
nights.

1. Input- season,
room type and
Stay nights
number.

- select rate
using nested
if based on
season and
room type
- total = rate * nights
- if night > 7
then apply 15%
discount
- Add total to
hotel revenue.

- final price
for guest
- Discount
amount
- Updated
hotel total
revenue

Pseudocode:

Begin

Input season (Peak or off Peak)

room type (standard, deluxe, suite)

no. of nights

If season = Peak then

If room type = standard then

rate = 5000

Else if room type = Deluxe then

rate = 8000

Else

rate = 12000

End if

Else

if room type = standard then

rate = 3000

Else if room type = deluxe

rate = 5000

Else

rate 8000

end if

end if

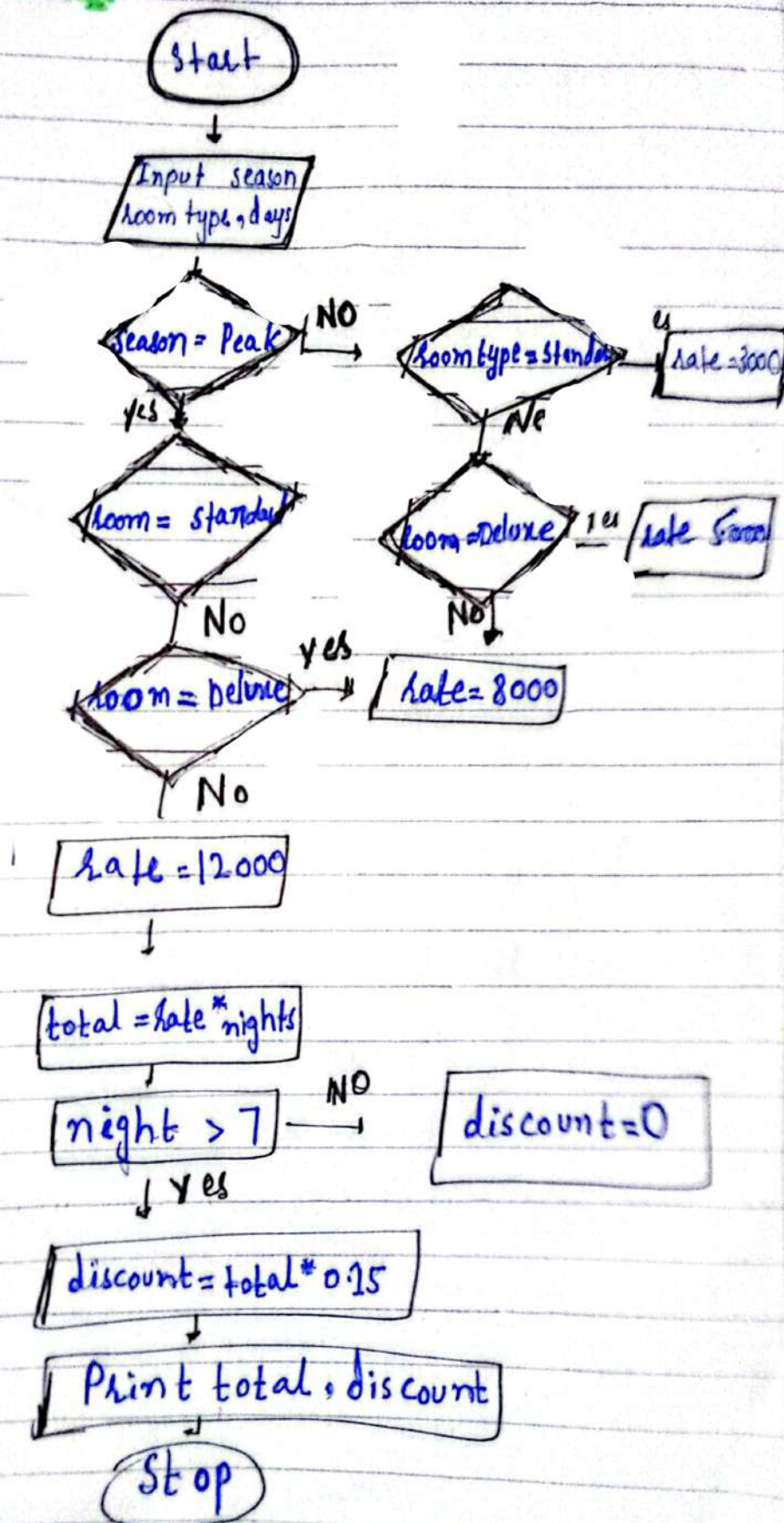
total = rate * nights

if night > 7 then

discount = total * 0.15

total = total - discount.

flow chart:



Q2: Elevator stimulation

Algorithm

Start

Set current = 0

Input list of N floor requests
for each floor in List DO

If floor > current then

print Moving up

Else if floor < current Then

print Moving down

Else

print Door Opening

End if

set current = floor

End for

stop .

Pseudocode

Begin

Current = 0

Input floor N

for i = 1 to n DO

if floor > Current Then

Output moving up

Else if floor < current Then

Output moving down

Else

Output door opening .

end

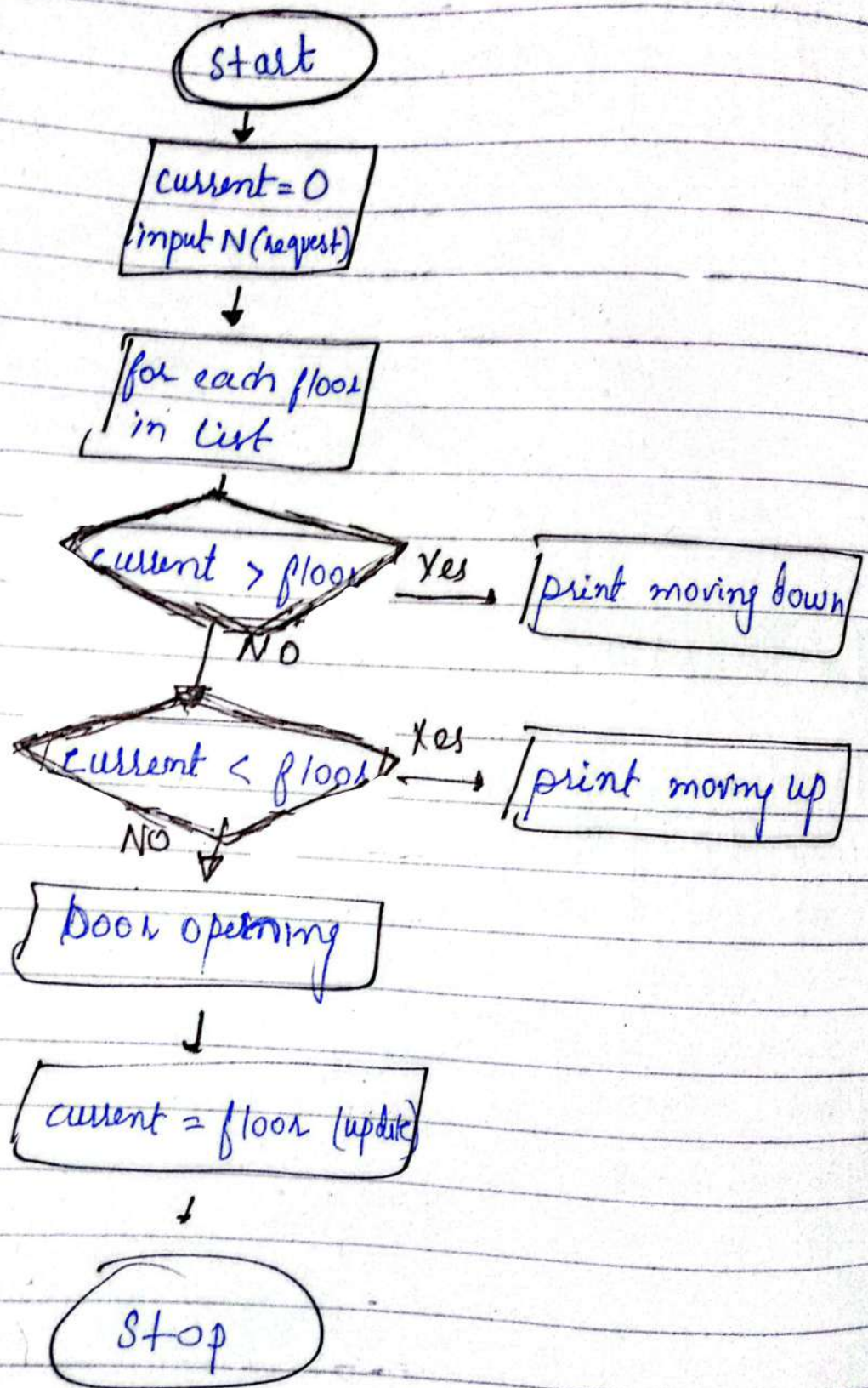
IPO Chart

Input	process	Output	Solution alternative
$N = \text{number of requests}$	• $\text{current} = 0$ • Compare floor with request • decide • $\text{update current} = \text{floor}$	• if $\text{current} > \text{request}$ moving down • $\text{current} < \text{request}$ moving up • else opening door	use a single diff variable insted of 2 comparisons. it reduces the number of condition.

PAC Chart

Problem	Input	Output	process
An elevator starts at floor 0 and receives N floor requests. for each request decide movement and update current floor	• Number of request	• Moving up if $\text{current} < \text{request}$ • Moving down if $\text{current} > \text{request}$ • $\text{current} = \text{request}$ opening door.	Set $\text{current} = 0$ Loop through each floor in list compare floor with current and print $\text{current} = \text{floor}$.

flow chart



Q3

Algorithm

Start

Input $N \leftarrow$ Number of student

For each students from 1 to N DO

Set $sum = 0$

has fail subject = False

for each subject from 1 to 5 DO .

Input mark

Set $sum = sum + mark$

if $mark < 33$ Then.

has fail subject = true

End if

End for

$average = sum / 5$

If has fail subject = true Then

print "fail - subject deficiency .

Else if $average \geq 80$ Then

print distinction

else if $average \geq 60$ Then

print pass

else

print fail

End if

End for

stop .

IPO chart:

Input	process	Output	solution alternative
Number of student	• Read 5 mark	• average ≥ 80	Use 2 flags instead of checking every mark separately.
Marks for each 5 subjects	• $sum = sum + Mark$	distinction	
	• Check if marks ≥ 33	• average $= 60$ pass	
	• average $= sum / 5$	• average < 60 fail	
	• Apply classification rules.		

PAC chart:

Problem	Input	Process	Output
Calculate average	Number of student	$sum = sum + mark$	classification according to average.
classify	Marks for each 5 subject	average $= sum / 5$	
distinction		if average ≥ 80	distinction / pass / fail
pass and fail.		distinction	
		average ≥ 60 pass	
		else fail	

Pseudocode:

Begin

Input N

for each student from 1 to N Do

Sum = 0

has fail subject $\leftarrow$ false

for $i \rightarrow 1$ to 5 Do

Input mark

Sum $\leftarrow$ sum + mark

if mark < 33 Then

has fail subject $\leftarrow$ True

End if

End for

average $\leftarrow$ sum / 5

If has fail subject = true Then

output fail $\leftarrow$ subject deficiency

else if average ≥ 80 Then

output Distinction

else if average ≥ 60 Then

Output average

else

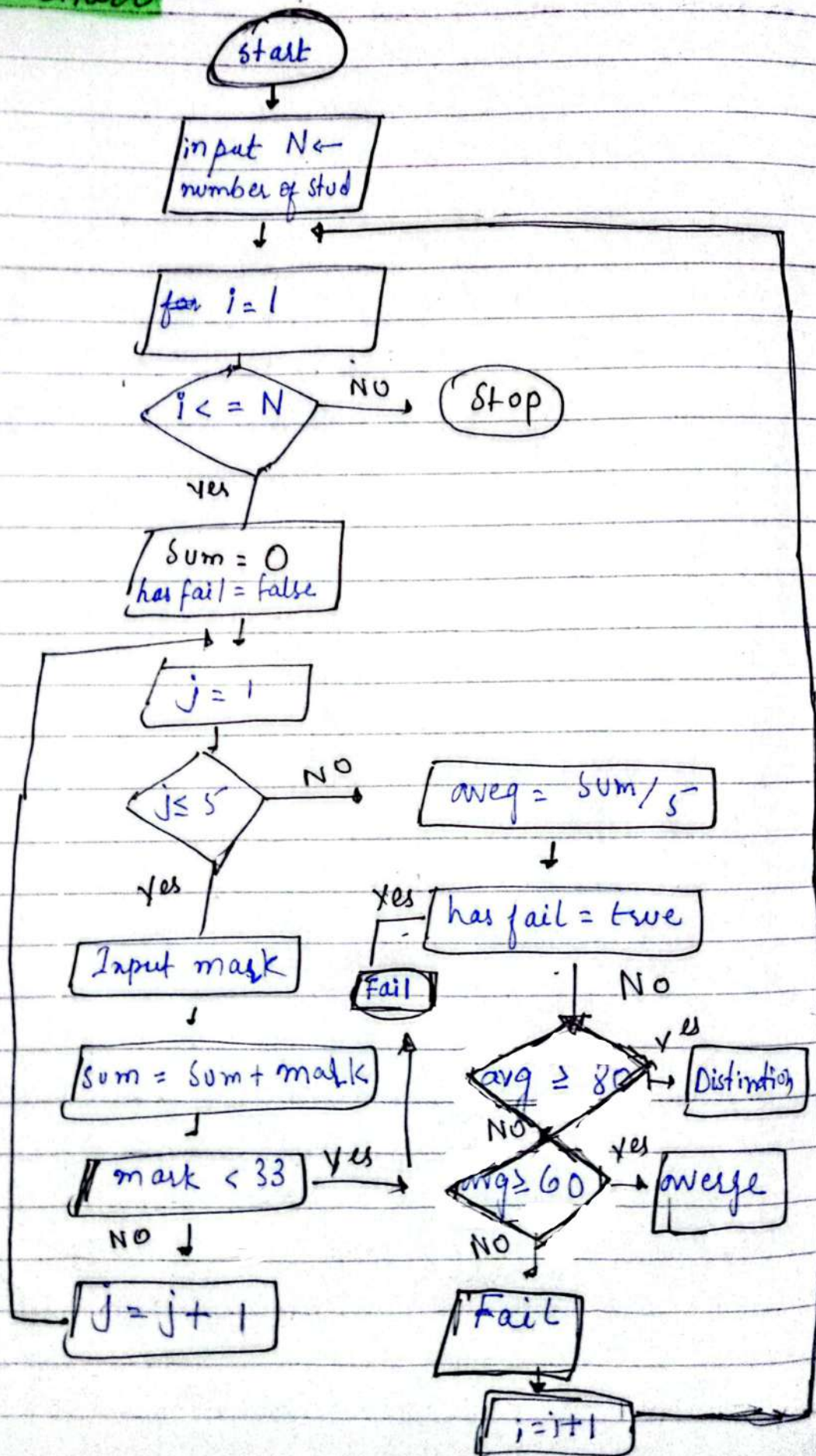
output fail

End if

End for

End

flow chart



Q4: Online shopping bill calculator

Algorithm

```
Input quantity
Input price per item
Input discount percentage
Input tax percentage
if  $q \leq 0$  OR  $p \leq 0$  OR  $d \leq 0$  OR  $t \leq 0$ 
    print "Error: Invalid Input".
end if
Set  $s = q * p$ 
Set  $a = s - (s * d) / 100$ 
Set final bill =  $a + (a * t) / 100$ 
    print bill details: s, a, final bill
Stop.
```

Pseudocode

Begin

```
Input q, p, d, t.
if  $q \leq 0$  OR  $p \leq 0$  OR  $d \leq 0$  OR  $t \leq 0$ 
    print Error: Invalid Input
end if
 $s \leftarrow q * p$ 
 $a \leftarrow s - (s * d) / 100$ 
final bill  $\leftarrow a + (a * t) / 100$ 
    out put Subtotal, s, Discount, final bill
end
```

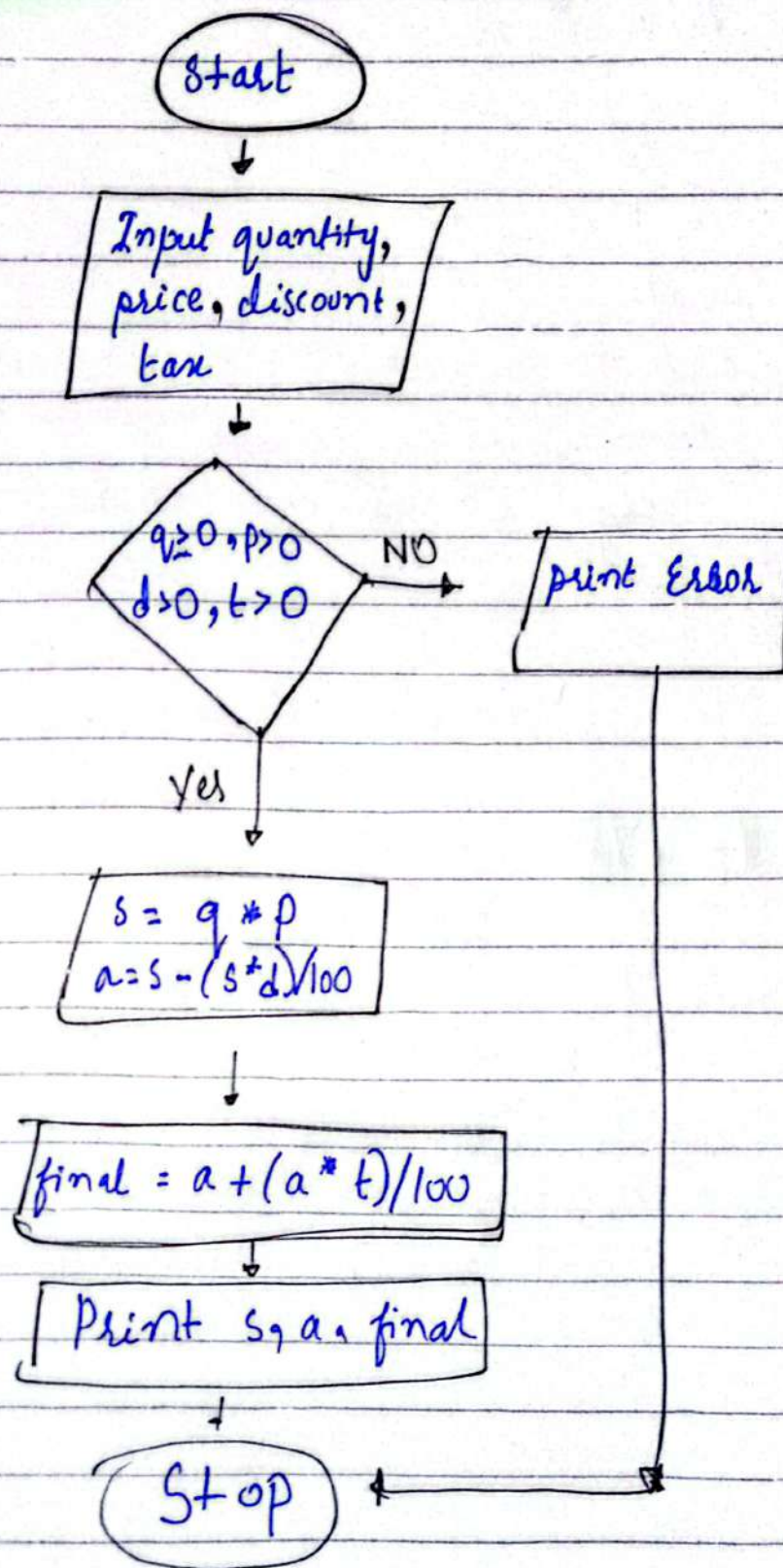
IPO chart:

Input	Process	Output	Solution alternative
- quantity - price per item - discount - Percentage - tax - percentage	$s = q * P$ $a = s - (s * d) / 100$ $final = a + (a * t) / 100$	- Subtotal bill - final bill - Invalid when $q < 0$ OR $P < 0$ OR $d < 0$ OR $t < 0$	Instead of doing all calculations in main, Use 3 separate functions. Each function calls the next one.

PAC Chart:

problem	Input	process	Output
calculate the final bill	- quantity - price per item - discount - percentage - tax percentage	- validate: $q \leq 0$, $P \leq 0$, $d \leq 0$, $t \leq 0$ - Then invalid $s = q * P$ $a = s - (s * d) / 100$ $final = a + (a * t) / 100$ - Store and display bill	- Subtotal bill - Discounted amount - final bill with tax.

flow chart



Q5

Algorithm

Start

Input N

for $i = 1$ to N

Repeat

Input vtype, user category, permit, emergency

If permit = Y or emergency = Y Then

Assign based on rules and capacity

End if

End for

Print summary

End.

Pseudocode

Begin

Input N

for $i = 1$ to N

Input vtype, user category, permit, emergency

If permit = Y Or emergency = Y Then

assign based on rules and capacity

End if

End for

Print summary

End.

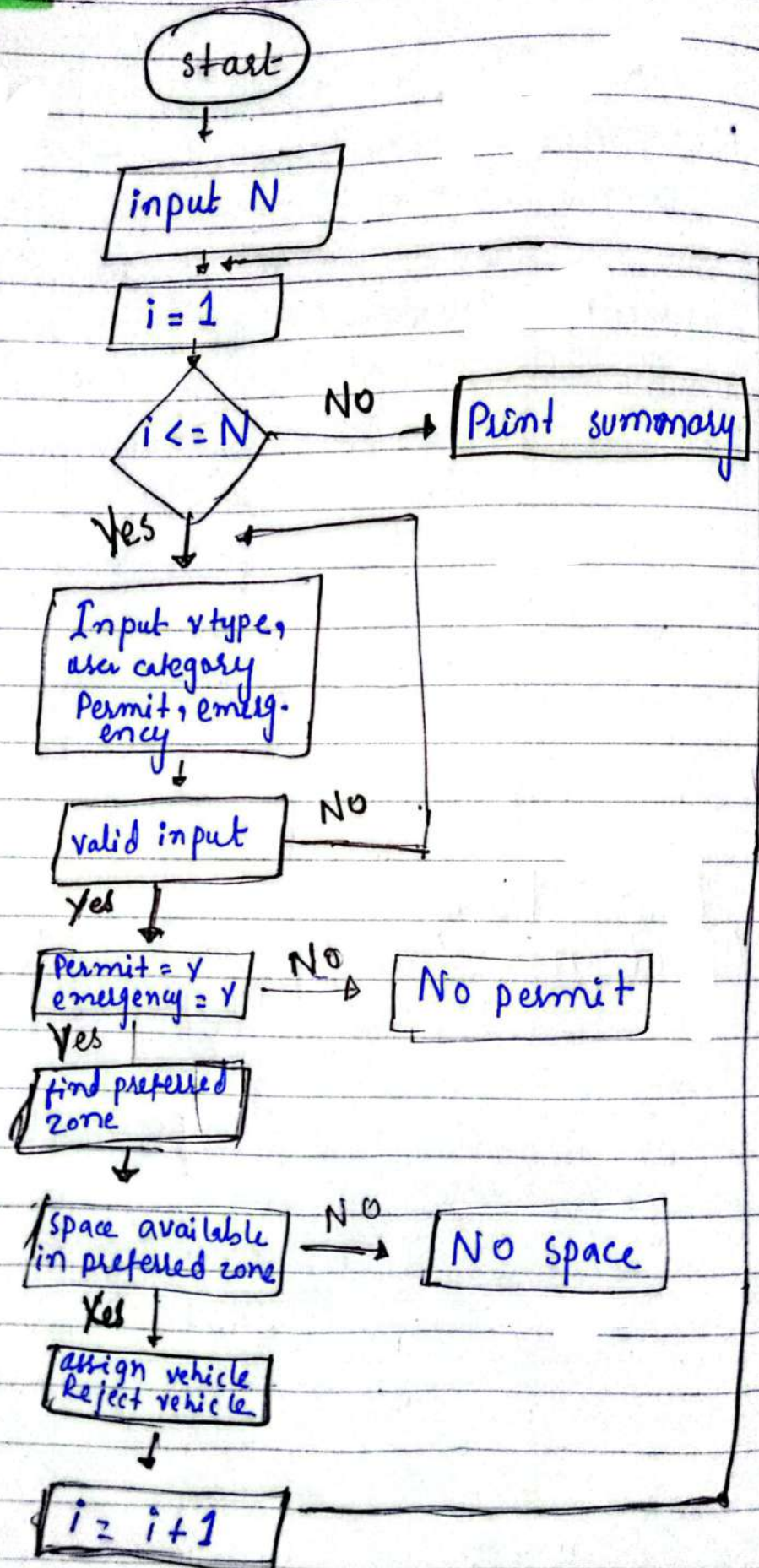
IPO chart

Input	Process	Output	Solution alternative
No. of vehicle user category permit emergency	- Input validation on Loop. - check permit/emergency - Determined preferred Zone - capacity check - assign and update counter	Zone assignment or rejection Total processed Count cars/Bikes/van final Zone occupancy	Use a Lookup table for Zone assignment. table will store preferred Zone and alternate Zone for each user type.

PAC chart

Problem	Input	Process	Output
assign to correct zone based on category	N, Vehicle-type, user category, permit, emergency	Check permit or emergency find preferred Zone check capacity assign or reject update.	Total accepted, Rejected vehicle counts, Zone occupancy, highest zone

flow chart



Q6 : Smart Ev charging and parking management system.

Algorithm

Start

Input vehicle type, battery Level, required Level, exp hours, current time.

Input membership disabled priority, station-available.

If stationavailable = N then

If vehicle type = H then

print charging unavailable - parking only.

else

print No charging slot available.

stop

End if

End if

If vehicle type = H then

Print vehicle does not qualify for EV charging.

Charging Allowed = false

Else if v type = E and battery Level $\geq$ 40 then

Print vehicle does not qualify for EV

Charging allowed = false

else

Charging allowed = true

End if

$\text{req charge} = \text{req Level} - \text{battery Level}$

If $\text{req charge} \leq 0$ Then
 print No charging required
 charging allowed = false
End if

If charging allowed = True Then

 If $\text{battery Level} \leq 15$ and $\text{req Level} \leq 80$ Then

 priority = Emergency charging priority

 Else if $\text{disable priority} = y$ or $\text{membership} = y$ AND $\text{battery Level} \leq 30$ Then

 priority = Normal charging

 End if

End if

If $\text{priority} \neq \text{Emergency charging priority}$
 AND $\text{membership} = y$ Then

 rate = $\text{rate} * 0.8$

End if

If charging allowed = true Then

 charging cost = $\text{req charge} * \text{rate}$

Else

 charging cost = 0

End if

If $\text{exp hours} \leq 2$ Then parking cost = 200

Else if $\text{exp hours} \leq 5$ Then parking cost = 400

Else parking cost = 700

If disabled priority = y then parking cost = 0
Else if membership = y then parking cost =
parking cost * 0.8

If exp hours > 8 then
warning = Long-stay warning:

Else

warning = standard parking duration.

End if

total = charging cost + parking cost

print "vehicle type", v type

print "current battery", battery Level, %

print "required charging", req charge, %

print "priority", priority

print "time status": time typ

print "Charging cost: Rs," charging cost

print "parking cost: Rs", parking cost

print "Final payable amount: Rs", total

print warning

Stop.

Pseudocode

Begin

Input vtype, battery level, req level, exp hours,
current time, membership, disabled priority,
station available

If station not available

If hybrid then parking only else no
slot AND stop

If hybrid OR Battery ≥ 40 Then No EV
charging.

req charge = req level - battery level

If recharge ≤ 0 then "No charging required"

If battery ≤ 15 and req ≥ 80 then
priority = Emergency

Else if disable = Y OR membership = Y AND
battery ≤ 30 then priority = priority

Else priority = Normal

Calculate charging cost based "on peak/off peak"
and membership.

calculate parking cost based on hours and
discounts

If hours > 8 Then Long-stay warning Else
standard

Display all results + total amount

End

IPO Chart

Input	process	Output	Solution altern-
v-type, battery level, req-level, exp-hours, current time, memb-disabled - priority, sation-available	• validate station-availability • check EV/hybrid, • calculate required charging Assign Priority • calculate charging cost with on-peak / Off-peak membership discount • calculate parking cost • check Cong stay warning	vehicle type current battery Required battery Priority time status charging cost parking cost final payable amount warning message.	Break the Logic into 3 functions: get priority, calculate charging calculate parking this makes the main algorithm cleaner instead of big nested structure.

PAC chart:

Problem	Input	process	Out put
Decide EV admission charging priority, calculate charging parking cost with discount and display summary.	v-type, battery Level, req Level, exp hours, current time, membership, disabled-priority, Station available.	check station check eligibility calculate req-charge Assign priority calculate costs Apply discounts generate-warning	vehicle-type, battery %, request charge%, priority, peak/off-peak, charging cost total warning message.

flow chart:

